

Feasibility study of Grid connected Photovoltaic system for electricity generation with peak demand deficit

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Abstract— Due to great interest in the development of renewable energy technologies in Morocco, this paper studied the performance of a solar photovoltaic grid-connected system located in Rabat, Morocco. Firstly a demand side management strategy has been developed to maximize energy efficiency of the system using a load shifting day-ahead approach. In the second part, an examination of resulted load requirement using a simulation model based on HOMER software is carried out. The analysis of the results has shown that the proposed model is an effective and feasible tool of electricity generation.

Keywords—Renewable energy; Smart Grid; Demand Side Management; Load shifting; Optimal generation.

I. INTRODUCTION

The global awareness of the need to reduce greenhouse emissions, and the increasing of global demand of energy brought an international environmental and energy debate. Morocco was among the first countries of the Middle East and North Africa to cut fossil fuel subsidies. It has abundant renewable energy resources, mainly solar, wind and hydro-power. In fact it has set targets to increase the share of electricity generating capacity from renewable to 42% by 2020 and 52% by 2030 as well as targets for reducing energy consumption to 12% by 2020 and 15% by 2030 through energy efficiency[1].

Recently, many studies have been made to supplement conventional generation with the grid connected solar photovoltaic (PV) systems which offers many benefits; Environmental by reducing carbon dioxide emissions (CO_2) and economic by minimizing the operation energy cost as presented in [2]. Though the introduction of these renewable technologies causes new challenges such as hard forecasted power and system stability. The Demand side management has become a hot topic to attack these obstacles and keep a permanent between the demand and power supply. The majority of studies present in the literature formulate the DSM

subject as an optimization problem with different strategies ,limits and conclusions as presented in [3,4,5,6]. Among many tools utilized for the optimization of hybrid systems; Hybrid Optimization of Multiple Electric Renewables (HOMER Pro Software) was carried out in this work due to its benefits. It allows a combination of various energy sources of both off-grid and grid-connected systems, has the ability of performing optimization and sensitive analysis, which gives it the ability of evaluating many possible system configurations easily and quickly. Therefore several studies have used it to examine the technical and economic feasibility of hybrid energy systems worldwide. The computation, simulation and optimization of a solar, Fuel cell and Biomass hybrid energy system was presented in [7]. In [8] an optimum sizing for rural electrification was performed. The work presented in [9] and [10] show the comparison of an optimized grid connected model and standalone grid with the objective of finding out the feasible model for the consumers. [11] focuses on the design of hybrid system based on PV-Biomass, Diesel connected to the grid and optimize the system configuration for different load profiles . The analyze of PV/diesel energy system performance under different conditions; including weather data, ambient temperature, solar radiation intensity and daily load profiles were applied for testing in [12]. In this context, the objective of this work is to study the feasibility and viability of grid connected solar PV system in Northern Africa (Rabat, Morocco). A load management strategy was applied in order to increase energy efficiency of the system by reducing the total load consumption. This first step has been achieved through an optimization program using an improved version of differential DE, Backtracking search Algorithm BSA which incorporates new developed elements, as utilization of dual population, one control parameter to control the amplitude of search direction and consulting historical population at each iteration in order to generate more efficient population [13]. Secondly an optimization of the distributed power, using HOMER program by introducing the load profile resulted by Demand

Side Management (DSM) strategy studied in the first part and predictive solar radiation intensity over a day is performed. In this context, this paper is organized as follows: Section 2 details the procedures of load management strategy. Section 3 describes the system studied and component's characteristics. Section 4 presents the results obtained by DSM strategy and the feasibility study of electricity generation. Then section 5 concludes the paper.

II. DEMAND SIDE MANAGEMENT METHODOLOGY

A. Problem formulation:

The proposed DSM strategy consists of load connection time shifting, in order to minimize the customer's peak load demand. A mathematical model can be given as follows:

$$f_1(t) : \sum_{t=1}^{24} [|P_{forecast}(t) - P_{T\ arg\ et}(t)| - |P_{load}(t)|] \quad (1)$$

where $P_{T\ arg\ et}$ is the target load demand curve. It is inversely proportional to the expected electricity prices. For example in an instant if the value of electricity prices is higher the value of target curve is low and vice-versa. The $P_{T\ arg\ et}$ is given by the following equation:

$$P_{T\ arg\ et}(t) = \left(\frac{C_{avg}}{C_{max}} \sum_{t=1}^{24} P_{forecast}(t) \right) \frac{1}{C(t)} \quad (2)$$

where C_{max} is the maximum price throughout the day of 24 hours C_{avg} is the average price during the day, and $C(t)$ the price of electricity at h^{th} hour. $P_{forecast}(t)$ is the forecasted hourly consumption at time t, which has been taken from the data of the smart grid mentioned in [14] including three areas i.e. residential, commercial and industrial. Further, $P_{load}(t)$ are the amounts of load to be connected or disconnected at each hour t depending on the variant ΔP expressed in equation (3).

$$\Delta P = P_{forecast} - P_{T\ arg\ et} \quad (3)$$

In order to approach the forecasted load profile as possible as to a target curve the variant ΔP is evaluated according to eq (4). If for an instant t the value of forecasted load is lower than the value of target load, the $P_{load}(t)$ is need to be added or connected. Wherever for an hour t the forecasted load is higher than the target load, $P_{load}(t)$ is subtracted or disconnected to the forecasted load at each instant t.

$$\begin{cases} \Delta P < 0, P_{load}(t) = Connect(t) \\ \Delta P > 0, P_{load}(t) = Disconnect(t) \end{cases} \quad (4)$$

$Connect(t)$ is the total amount of load which needs to be connected at hour h so that the valley could be filled up. It is calculated at those instants when the value of ΔP is negative. At other instants the value of connected load is zero. The

connection moments are usually the ones having low prices. The $Connect(t)$ is given by the equation (5):

$$connect(t) = \sum_{k=1}^D X_{kh} P_{1k} + \sum_{j=1}^{l-1} \sum_{k=1}^D X_{k(h+j)} P_{(1+j)k} \quad (5)$$

The equation (5) is composed of two parts. The first part calculates the power consumption by the X number of devices at h^{th} hour. In the second part, if the device is having more than one operating hours, power consumption is calculated at successive hours, i.e. $h+1, h+2, \dots, (h+l-1)$.

X_{kh} and $X_{k(h+l)}$ is the quantity of devices of type k assumed to be operating at hour and ; D is the number of device type, P_{1k} is the power consumption of device k in the first hour of operation and $P_{(1+j)k}$ is the power consumption of the same device at subsequent hours.

$$Disconnect(t) = \sum_{k=1}^D X_{kh} P_{1k} + \sum_{j=1}^{l-1} \sum_{k=1}^D X_{k(h+j)} P_{(1+j)k} \quad (6)$$

$Disconnect(t)$ is the total amount of load which needs to be disconnected from h^{th} hour, so at that hour our loads must be reduced. It is calculated for those instants which have positive value of ΔP . At other instants the value of disconnected load is zero. $Disconnect(t)$ is calculated through the equation(6). The terms in this equation state the same meaning to those of equation (5).

Constraints:

Due to the operational limitations of the system, the following constraints must be satisfied throughout the operation of the system for any feasible solution.

— The number of devices to be shifted at any time step can not be more than the sum of all the controllable devices controlled at that time instant.

$$\sum_{t=1}^{24} X_{kh} \leq controllable(k) \quad (7)$$

where $controllable(k)$ is the sum of devices of type k available for the control.

— Number of devices shifted away from a time step, can not take a negative value.

$$X_{kh} \succ 0, \forall k, h \quad (8)$$

B. Proposed DSM algorithm

In the objective of planning, monitoring and implementing activities aiming to reduce total costs of meeting energy demand, a load shifting technique is proposed at the first step of this work, as shown in Fig.2. It takes advantage of time independence of loads, and shifts loads from peak time to off-peak periods, and presents the most effective load management technique in current distribution networks [15]. In several studies DSM strategies were considered as an optimization problem in order to minimize the fossil fuel consumption,

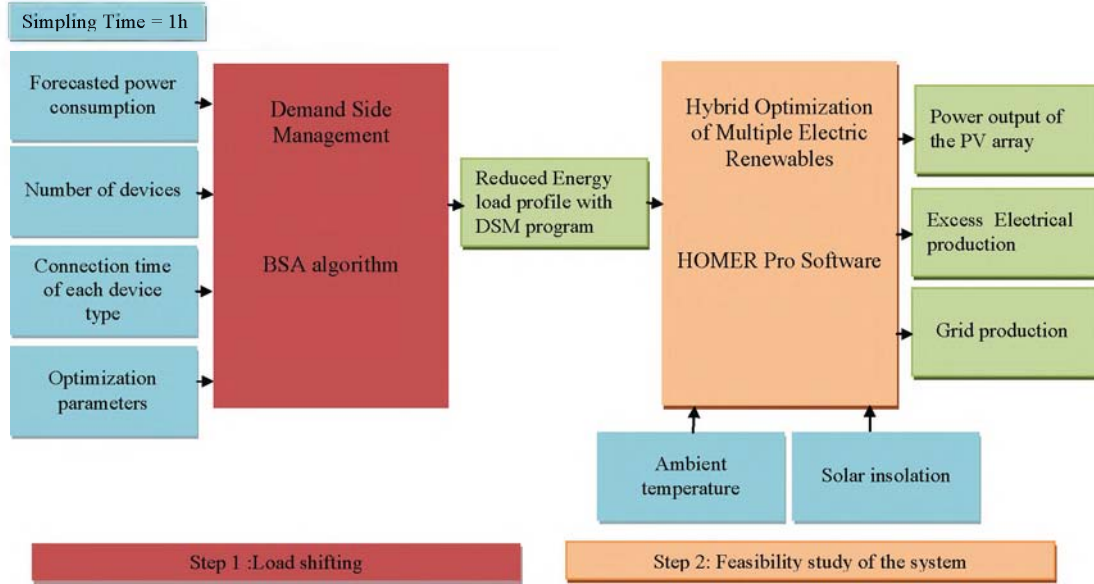


Fig. 1. Overview of the proposed study

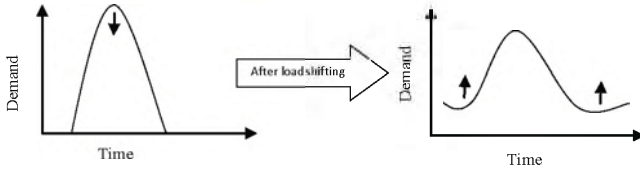


Fig. 2. Load shifting approach

peak load profile, total cost energy of the system, maximize the renewable energy utilization (solar PV and Wind...) or users comfort coefficient.

Optimization methods can be classified in different ways depending on three different characteristics [16]. Firstly, according to user's interactions; individual or cooperative users, then the optimization approach utilized stochastic or deterministic and thirdly based on time scale characteristic if it's a day ahead timing or a real time. In this work an effective algorithm BSA based on day ahead load shifting management technique is used to minimize the objective function equation (1). The procedure of BSA can be defined in five processes: initialization, selection-I, mutation, crossover and selection-II [17]. Fig.3 presents the main steps of the proposed load scheduling strategy. With the aim of better understanding the effect of load time shifting strategy, a typical day has been controlled from the 8AM hour of the current studied day to the eighth hour of the next day.

III. ENERGY SYSTEM MANGEMENT OF THE PRROPOSED SYSTEM

A. Load profile

In this study, the hybrid system is designed for the electrification of three different areas (residential, commercial,

and industrial), in Rabat, Morocco (latitude 33°58,3'N and longitude 6°51,0'W). The load profile of each customer type

Start

1-**function BSA** (Objective function, Dimension rate, Population size)

2-**Inputs data from utility:**

Forecasted load, Electricity price, Percentage, Device parameters (Power consumption, Connexion time, total number...)

While iteration <= 100

3-**Target curve calculation**

$$P_{\text{Target}}(t) = \left(\frac{C_{\text{avg}}}{C_{\text{max}}} \sum_{t=1}^{24} P_{\text{forecast}}(t) \right) \frac{1}{C(t)}$$

4-**Initialization of population**

5- **Initialization of old population**

6-**Initialization of objective function** //for each initial population value.

7-**Generation of trial population**

8-**Objective function evaluation**

// (objective function =Best fitness)

If best fitness (i)=min(objective(i)

Update the best fitness

Else

Keep the previous fitness

9-**Outpouts algorithm:**

%Load peak reduction

$$P = \frac{\text{reduced demand}}{\text{Max demand}}$$

End.

End.

Fig. 3. Main steps of DSM strategy

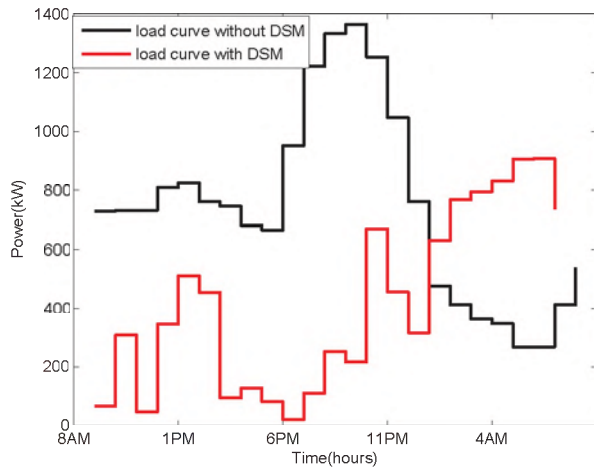


Fig. 4. DSM for Residential users

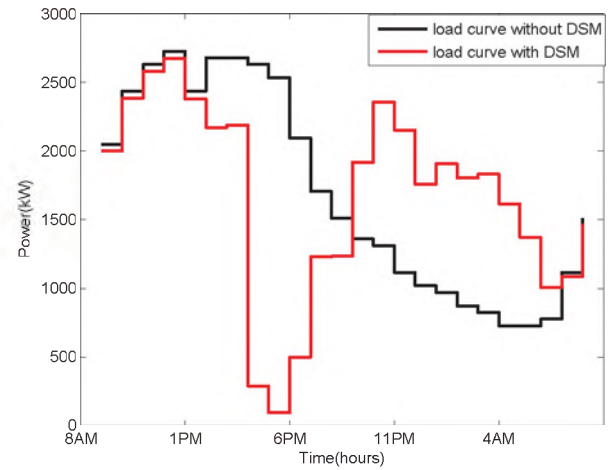


Fig. 6. DSM for Industrial users

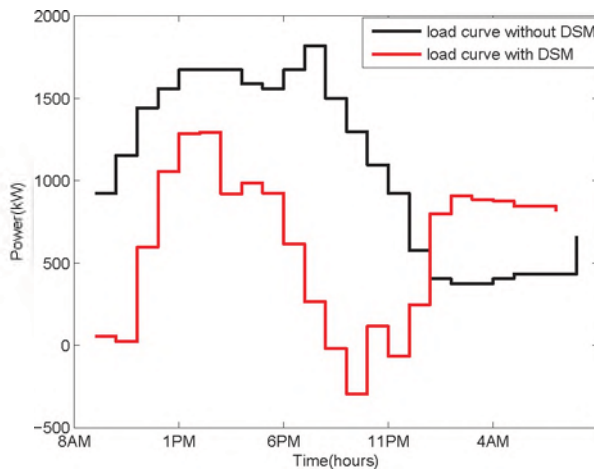


Fig. 5. DSM for Commercial users

has been taken from DSM strategy outputs, results of the first part of this work, and fed into HOMER as a daily profile, and also used as a power balance constraint. As presented in Fig.4, a prominent peak demand period occur late in the evening, from 6PM until 11PM for residential service customers what can be explained by usual evening activities (cooking, lighting, TV, reading). Fig.5 depicts the commercial load profile, it can be observed two peak demand periods first one in the afternoon between 1PM and 6PM then at night from 2AM to 6AM, while consumption in the industrial zone has a wide peak period of 10 hours, due to long hours of work of the industry as shown in Fig.6.

B. Energy system components

The proposed energy system consists of PV panels and national electricity grid system. The schematic diagram of the grid-connected PV system is shown in Fig.6. Detailed descrip-

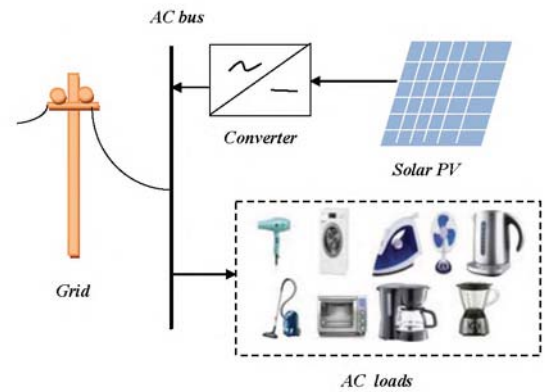


Fig. 7. The schematic diagram of PV and grid-connected system

TABLE I
COMPONENTS CHARACTERISTICS

Components	Total		
	Residential Users	Commercial Users	Industrial Users
PV Generic flat plate (Kwh/yr)	6 238 083	8 911 548	26 734 643
Battery Generic Lead Acid (Strings)	2	2	9
Converter system (Kwh/yr)	5 640 946	8 230 242	9 213 252
Grid (Kwh/yr)	2 599 361	3 836 064	24 574 080

tions and characteristics of each component are presented in Table I.

1) *Solar PV modules*: Generic flat plate PV has been used in this system to generate electricity from solar energy. The PV array with a rated power of 3500kW was used for residential users, 5000 kW for commercial consumers, and 15000kW for industrial users. To ensure more usage of RES we have kept the minimum renewable fraction at 50% as a constraint for the optimization problem.

2) *Grid*: In this designed model, grid is used to supply electricity when the renewable energy power is lower than the total consumer demand.

3) *Battery*: Though renewable resources provide an average amount of energy for a long period of time, the storage unit is used to supply the consumer demand when the resources provide a low amount of energy. We have used Generic Lead Acid energy storage units to store the energy supply at a nominal voltage of 12 V.

4) *Converter*: Converter can be used to convert AC to DC and vice versa which means that it can be used both as rectifier and inverter [9]. In the proposed hybrid system the converter is applied to convert the DC voltage supplied by PV module and batteries as the supplied energy is used by AC load.

IV. RESULTS DISCUSSION

A. Peak load reduction

To evaluate the usefulness of the proposed algorithm the simulations were repeated 100 times with a number of population of 20 and 0,7 as dimension rate. We carried out the optimization in three different areas containing residential loads with 16 different equipment's type, commercial loads (808 devices) with the highest power consumption and industrial loads with the least total device number(109 equipments for 8 different types)in Rabat, region of Morocco. From the analysis of DSM implementation, it can be concluded that on reshaping from peak hours to off peak periods, has significantly reduced, for various types of consumers. The obtained results for residential customers, presented in Fig.4 shows a reduction of utility load of 33% from 1336,6 kWh to 907,34 kWh. In Fig.5, the peak load curve in commercial area has been reduced from 1818,2 kWh to 1286,1 kWh, which presents a minimization factor of 29 %, while the simulation results of industrial customers shows a minimization of total energy consumption of 24% from 2727,7 kWh without DSM to 2278,6 kWh using the proposed DSM technique as presented in Fig.6. Peak demand power reductions due to demand side management strategy are summarized graphically in Fig.8.

B. Electricity Generation

In the second part of this work, the software HOMER Pro has been used as the sizing and optimization software tool. The simulation was performed in order to obtain the optimum electricity generation that meets the previously resulted load profiles from the DSM strategy studied firstly. Grid-connected solar PV system has been analyzed, during one year. Hourly simulation was performed to evaluate the feasibility of the system, with different consumer profiles (Residential,Commercials, and Industrials), located in Rabat, Morocco. Typical daily electricity production of each source is shown in Fig.9, Fig.10 and Fig.11. An uniform load about 20375 kWh per day is observed in the residential area.The major amount about 70,6% of the total electricity generated is provided by the renewable source(PV),while the grid ranks second by 29,4% (6664kWh/day).

Fig.10 presents the hourly electricity generation for commercial users . It can be noticed that an important fraction of electricity was generated from PV generator about 29082 kWh/day which presents 69,9%. However, 30,1% (9720 kWh/day) was shared from grid electricity.

Fig.11 indicates the same hourly data but for the industrial area. It can be observed that the renewable energy contribution exceeds the electricity grid production.

Hence, it can be further concluded from these results that the proposed power system is feasible and good solution to mitigate pollutant emissions from conventional power sources.

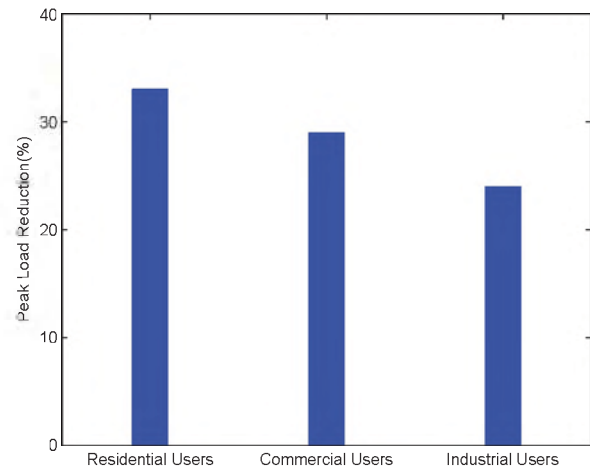


Fig. 8. Peak load reduction.

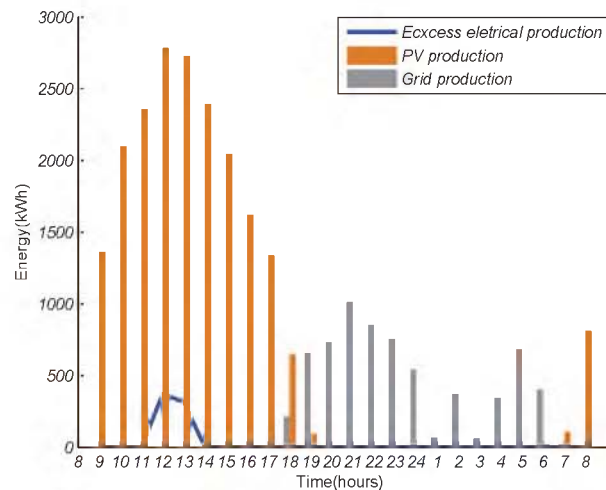


Fig. 9. Daily distribution of electricity generation in Residential area.

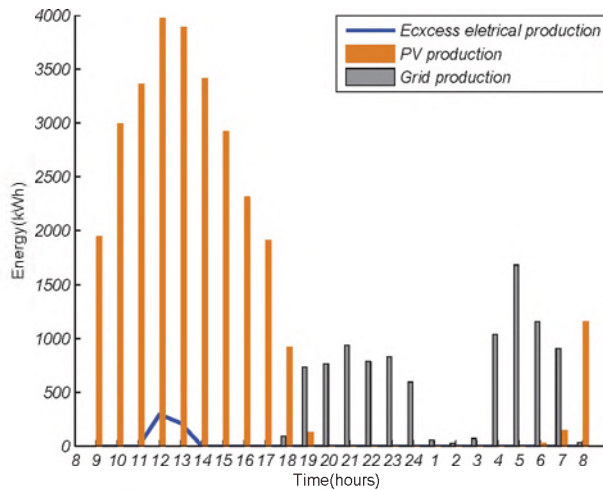


Fig. 10. Daily distribution of electricity generation in Commercial area.

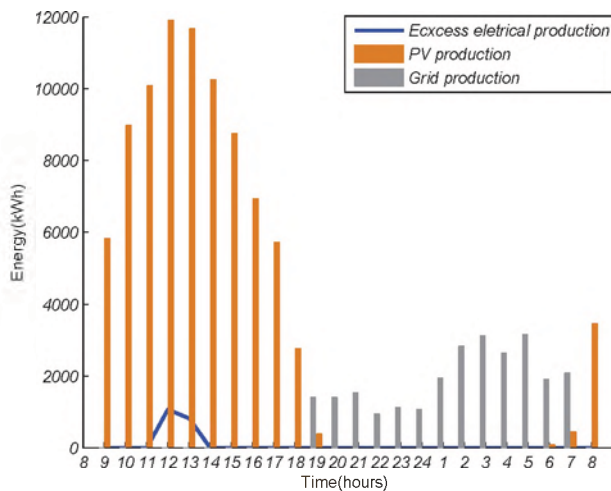


Fig. 11. Daily distribution of electricity generation in Industrial area.

V. CONCLUSION

This paper examines the feasibility and viability of Grid-connected PV system for an optimal electricity generation in Rabat, Morocco. A first stage was focused on minimization of peak demand curve using BSA. Then as a second step the resulted load profiles were fed into HOMER software. The analysis of the results has shown that renewable energy source (PV) is able to ensure the high fraction of power demand in the three areas. Therefore, the proposed and optimized system is a feasible and effective solution for electrical power production.

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